LAPORAN PRAKTIMUM MYSQL / MARIA DB

IMPLEMENTASI SQL LEFT JOIN, RIGHT JOIN, dan DDL/DML VIEW

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Aktifitas : Fullstack Web Dev | Tugas : Database/mysql 19/10/26

Link General Github : https://github.com/jaraa4/Tugas_SIB4_FWD.git

Link Tugas : https://github.com/jaraa4/Tugas_SIB4_FWD/tree/main/Database/database%204/db_tugas_join

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1. **Pendahuluan & Skema Tabel Contoh**

   Laporan ini membahas penggunaan operasi penggabungan tabel LEFT JOIN dan RIGHT JOIN pada RDBMS, serta pembuatannya dalam bentuk objek VIEW. Kasus yang digunakan mengacu pada basis data akademik dan penjualan retail.

   a. **Struktur Tabel Kasus 1 (Sistem Akademik):**

      - **Mahasiswa (id_mahasiswa, nama_mahasiswa, prodi)**
      - **KRS (id_krs, id_mahasiswa, nama_matakuliah, sks)**

2. **Implementasi Left Join – Mahasiswa & Pengambilan KRS**

   a. **Membuat Database Terlebih dahulu**

```
MariaDB [(none)]> CREATE DATABASE db_tugas_join;
Query OK, 1 row affected (0.015 sec)

MariaDB [(none)]>
```

   b. Menggunakan database yang di buat

```
MariaDB [(none)]> USE db_tugas_join;
Database changed
MariaDB [db_tugas_join]>
```

   c. Membuat table mahasiswa

```
MariaDB [db_tugas_join]> CREATE TABLE Mahasiswa (
    ->     id_mahasiswa INT PRIMARY KEY,
    ->     nama_mahasiswa VARCHAR(50),
    ->     prodi VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.027 sec)
```

   d. Membuat table krs

```
MariaDB [db_tugas_join]> CREATE TABLE KRS (
    ->     id_krs INT PRIMARY KEY,
    ->     id_mahasiswa INT,
    ->     nama_matakuliah VARCHAR(50),
    ->     sks INT
    -> );
Query OK, 0 rows affected (0.017 sec)
```

   e. Memasukan data sample ke table mahasiswa

```
MariaDB [db_tugas_join]> INSERT INTO Mahasiswa VALUES
    -> (1, 'Zahra Zaskia', 'Teknik Informatika'),
    -> (2, 'Ahmad Fauzi', 'Sistem Informasi'),
    -> (3, 'Siti Nurhaliza', 'Sistem Informasi');
Query OK, 3 rows affected (0.090 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

   f. Menambahkan data sample ke table krs

```
MariaDB [db_tugas_join]> INSERT INTO KRS VALUES
    -> (101, 1, 'Basis Data', 3),
    -> (102, 1, 'Pemrograman Web', 3);
Query OK, 2 rows affected (0.018 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

g. Membuat table pelanggan dan pesanan

```
MariaDB [db_tugas_join]> CREATE TABLE Pelanggan (
    ->     id_pelanggan INT PRIMARY KEY,
    ->     nama_pelanggan VARCHAR(50),
    ->     kota VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.019 sec)

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> CREATE TABLE Pesanan (
    ->     id_pesanan INT PRIMARY KEY,
    ->     id_pelanggan INT,
    ->     total_belanja INT
    -> );
Query OK, 0 rows affected (0.012 sec)
```

h. Menambahkan data sample ke table pesanan dan pelanggan

```
MariaDB [db_tugas_join]> INSERT INTO Pelanggan VALUES
    -> (1, 'Budi Santoso', 'Jakarta'),
    -> (2, 'Rina Wijaya', 'Bogor'),
    -> (3, 'Dewi Lestari', 'Depok');
Query OK, 3 rows affected (0.019 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> INSERT INTO Pesanan VALUES
    -> (1001, 1, 150000),
    -> (1002, 2, 250000);
Query OK, 2 rows affected (0.004 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

3. Implementasi Right Join – Dosen / Mata Kuliah

a. Membuat table dosen dan mata kuliah

```
MariaDB [db_tugas_join]> CREATE TABLE Dosen (
    ->     id_dosen INT PRIMARY KEY,
    ->     nama_dosen VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.015 sec)

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> CREATE TABLE MataKuliah (
    ->     id_mk INT PRIMARY KEY,
    ->     nama_mk VARCHAR(50),
    ->     id_dosen INT
    -> );
Query OK, 0 rows affected (0.014 sec)
```

b. Menambahkan data sample ke table dosen dan mata kuliah

```
MariaDB [db_tugas_join]> INSERT INTO Dosen VALUES
    -> (1, 'Dr. Aris Purwanto'),
    -> (2, 'Prof. Endang');
Query OK, 2 rows affected (0.008 sec)
Records: 2  Duplicates: 0  Warnings: 0

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> INSERT INTO MataKuliah VALUES
    -> (201, 'Algoritma & Struktur Data', 1),
    -> (202, 'Jaringan Komputer', NULL);
Query OK, 2 rows affected (0.008 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

c. Membuat table supplier dan produk

```
MariaDB [db_tugas_join]> CREATE TABLE Supplier (
    ->     id_supplier INT PRIMARY KEY,
    ->     nama_supplier VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.013 sec)

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> CREATE TABLE Produk (
    ->     id_produk INT PRIMARY KEY,
    ->     nama_produk VARCHAR(50),
    ->     stok INT,
    ->     id_supplier INT
    -> );
Query OK, 0 rows affected (0.009 sec)
```

d. Menambahkan data sample ke table supplier dan produk

```
MariaDB [db_tugas_join]> INSERT INTO Supplier VALUES
    -> (1, 'PT Mitra Utama');
Query OK, 1 row affected (0.009 sec)

MariaDB [db_tugas_join]>
MariaDB [db_tugas_join]> INSERT INTO Produk VALUES
    -> (301, 'Laptop Asus', 10, 1),
    -> (302, 'Mouse Wireless', 25, NULL);
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

4. Eksekusi Kueri JOIN & CREATE VIEW

a. LEFT JOIN (Mahasiswa & KRS)

```
MariaDB [db_tugas_join]> SELECT m.id_mahasiswa, m.nama_mahasiswa, m.prodi, k.nama_matakuliah, k.sks
    -> FROM Mahasiswa m
    -> LEFT JOIN KRS k ON m.id_mahasiswa = k.id_mahasiswa;
+--------------+----------------+-------------------+-----------------+------+
| id_mahasiswa | nama_mahasiswa | prodi             | nama_matakuliah | sks  |
+--------------+----------------+-------------------+-----------------+------+
|            1 | Zahra Zaskia   | Teknik Informatika | Basis Data      |    3 |
|            1 | Zahra Zaskia   | Teknik Informatika | Pemrograman Web |    3 |
|            2 | Ahmad Fauzi    | Sistem Informasi  | NULL            | NULL |
|            3 | Siti Nurhaliza | Sistem Informasi  | NULL            | NULL |
+--------------+----------------+-------------------+-----------------+------+
4 rows in set (0.014 sec)
```

b. Membuat view

```
MariaDB [db_tugas_join]> CREATE VIEW view_left_join_krs AS
    -> SELECT m.id_mahasiswa, m.nama_mahasiswa, m.prodi, k.nama_matakuliah, k.sks
    -> FROM Mahasiswa m
    -> LEFT JOIN KRS k ON m.id_mahasiswa = k.id_mahasiswa;
Query OK, 0 rows affected (0.011 sec)
```

c. Cek view

```
MariaDB [db_tugas_join]> SELECT * FROM view_left_join_krs;
+--------------+----------------+--------------------+-----------------+------+
| id_mahasiswa | nama_mahasiswa | prodi              | nama_matakuliah | sks  |
+--------------+----------------+--------------------+-----------------+------+
|            1 | Zahra Zaskia   | Teknik Informatika | Basis Data      |    3 |
|            1 | Zahra Zaskia   | Teknik Informatika | Pemrograman Web |    3 |
|            2 | Ahmad Fauzi    | Sistem Informasi   | NULL            | NULL |
|            3 | Siti Nurhaliza | Sistem Informasi   | NULL            | NULL |
+--------------+----------------+--------------------+-----------------+------+
```

d. LEFT JOIN (Pelanggan & Pesanan)

```
MariaDB [db_tugas_join]> SELECT p.id_pelanggan, p.nama_pelanggan, p.kota, ps.id_pesanan, ps.total_belanja
    -> FROM Pelanggan p
    -> LEFT JOIN Pesanan ps ON p.id_pelanggan = ps.id_pelanggan;/
+--------------+----------------+---------+------------+----------------+
| id_pelanggan | nama_pelanggan | kota    | id_pesanan | total_belanja  |
+--------------+----------------+---------+------------+----------------+
|            1 | Budi Santoso   | Jakarta |       1001 |         150000 |
|            2 | Rina Wijaya    | Bogor   |       1002 |         250000 |
|            3 | Dewi Lestari   | Depok   |       NULL |           NULL |
+--------------+----------------+---------+------------+----------------+
```

e. Membuat view

```
MariaDB [db_tugas_join]> CREATE VIEW view_left_join_pesanan AS
    -> SELECT p.id_pelanggan, p.nama_pelanggan, p.kota, ps.id_pesanan, ps.total_belanja
    -> FROM Pelanggan p
    -> LEFT JOIN Pesanan ps ON p.id_pelanggan = ps.id_pelanggan;
Query OK, 0 rows affected (0.004 sec)
```

f. Cek view

```
MariaDB [db_tugas_join]> SELECT * FROM view_left_join_pesanan;
+--------------+----------------+---------+------------+----------------+
| id_pelanggan | nama_pelanggan | kota    | id_pesanan | total_belanja  |
+--------------+----------------+---------+------------+----------------+
|            1 | Budi Santoso   | Jakarta |       1001 |         150000 |
|            2 | Rina Wijaya    | Bogor   |       1002 |         250000 |
|            3 | Dewi Lestari   | Depok   |       1003 |         175000 |
+--------------+----------------+---------+------------+----------------+
3 rows in set (0.001 sec)
```

5. Right join dosen dan mata kuliah

a. Right Join

```
MariaDB [db_tugas_join]> SELECT * FROM view_right_join_matkul;
+-------------------+-------+---------------------------+
| nama_dosen        | id_mk | nama_mk                   |
+-------------------+-------+---------------------------+
| Dr. Aris Purwanto |   201 | Algoritma & Struktur Data |
| Prof. Endang      |   202 | Jaringan Komputer         |
+-------------------+-------+---------------------------+
2 rows in set (0.002 sec)
```

b. Buat view

```
MariaDB [db_tugas_join]> CREATE OR REPLACE VIEW view_right_join_matkul AS
    -> SELECT d.nama_dosen, mk.id_mk, mk.nama_mk
    -> FROM Dosen d
    -> RIGHT JOIN MataKuliah mk ON d.id_dosen = mk.id_dosen;
Query OK, 0 rows affected (0.009 sec)
```

c. cek view

```
MariaDB [db_tugas_join]> SELECT * FROM view_right_join_matkul;
+-------------------+-------+---------------------------+
| nama_dosen        | id_mk | nama_mk                   |
+-------------------+-------+---------------------------+
| Dr. Aris Purwanto |   201 | Algoritma & Struktur Data |
| Prof. Endang      |   202 | Jaringan Komputer         |
+-------------------+-------+---------------------------+
2 rows in set (0.002 sec)
```